

- 7 (a) due to the lost volts in internal resistance/cell or energy losses in the internal resistance/cell B1 [1]
- (b) (i) $V = IR$ C1
- $= 1.2 \times 6 = 7.2 \text{ V}$ A1 [2]
- (ii) p.d. across Y and internal resistance $r = 4.8 \text{ (V)}$ [12 – 7.2] C1
- resistance of Y + $r = 4.8 / 1.2 = 4 \text{ (}\Omega\text{)}$ C1
- resistance of Y = $4 - 0.5 = 3.5 \text{ }\Omega$ A1 [3]
- or
- $R_{\text{total}} = 12 / 1.2 = 10 \text{ (}\Omega\text{)}$ (C1)
- $X + r = 6.5 \text{ (}\Omega\text{)}$ (C1)
- resistance of Y = $3.5 \text{ }\Omega$ (A1)
- (iii) $P = I^2 r$ C1
- $= (1.2)^2 \times 0.5 = 0.72 \text{ W}$ A1 [2]
- (c) terminal p.d. increases as R is increased
current decreases so there are less lost volts B1 [1]